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Semiclassical transition state theory based on fourth order vibrational perturbation theory: Model system studies beyond symmetric Eckart barrier

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The following one-dimensional model potential barriers are studied to compute the tunneling probabilities using semiclassical transition state theory (SCTST) based on fourth order vibrational perturbation theory (VPT4): (i) Morse barrier, (ii) asymmetric Eckart barrier, (iii) adjacent well barrier, and (iv) first order approximation to the symmetric Eckart barrier. The current investigation extends a previous study performed for the symmetric Eckart barrier. The performance of VPT4-SCTST is investigated for a broad range of energies, including the very deep tunneling regime (approximately 1% of the barrier height), and the results are compared to the exact quantum mechanical treatment, as well as to the popular second order vibrational perturbation theory (VPT2)-SCTST approach. While the VPT4-SCTST results are in excellent agreement with their quantum mechanical counterparts for a wide range of energies above and below the barrier, it is found for systems (ii) and (iv) that higher order terms (in $n + \frac{1}{2}$) in energy expansion are important for similarly excellent agreement in the deep tunneling regime. For the asymmetric Eckart barrier, the convergence properties of SCTST are analyzed in terms of its analytically known Siegert eigenvalues. A simple empirical modification of VPT2-SCTST advocated in the aforementioned previous work for the symmetric Eckart barrier is also tested here for the new model systems, and it is shown not to provide a uniformly systematic improvement. *Published by AIP Publishing.* <https://doi.org/10.1063/1.5040978>

I. INTRODUCTION

A rigorous treatment of tunneling in reaction rate theory has been a longstanding challenge in chemical physics. A full state-to-state quantum reactive scattering calculation is exact and certainly accounts for tunneling. However, not only such a calculation is computationally demanding but also it does not have the simplicity of a theory such as transition state theory (TST). One solution to this problem is provided by the reactive flux correlation function based formalism of computing formally exact quantum mechanical rates, which was pioneered by Miller *et al.*¹ A central quantity in this formulation is the cumulative reaction probability (CRP), $G(E)$, from which the microcanonical and canonical rate constants can be obtained as follows:

$$\begin{aligned} k(E) &= (2\pi\hbar\rho_r(E))^{-1}G(E), \\ k(T) &= (2\pi\hbar Q_r(T))^{-1} \int_{-\infty}^{\infty} dE e^{-\beta E} G(E). \end{aligned} \quad (1)$$

Given that one has access to the density of state $\rho_r(E)$ or partition function $Q_r(T)$ for the reactant, the focus becomes the evaluation of CRP. The CRP can also be thought of as the “number of states which react” and can be expressed as

$$G(E) = \sum_{n_r} P_{n_r}(E), \quad (2)$$

where $P_{n_r}(E)$ is the total reaction probability from state n_r . A number of methods have been developed for an efficient evaluation of the CRP^{2–5} (or its canonical counterpart,

the flux-flux correlation function^{6–8}), and many reactions have been studied using these techniques.^{9–11} Despite its success, reactive flux correlation function based methods tend to be computationally expensive and often require the knowledge of a global reactive potential energy surface. This limits the applicability of these methods to rather small systems.

An alternative *semiclassical* approach to compute reaction rate constants was developed by Miller a few decades ago, known as semiclassical transition state theory (SCTST).¹² It has the form of a generalized TST as the non-separability of the Hamiltonian is taken into account. As SCTST is often used in conjunction with vibrational perturbation theory (VPT), in principle it is systematically improvable. In addition, it bypasses the need for an optimized dividing surface, as the energy is always expanded around the transition state geometry, which is taken to be the reference point. These features make SCTST an attractive alternative to the more demanding fully quantum rate theories. The underlying idea behind SCTST (and its VPT variants) is that the classical Hamiltonian can be expressed in terms of good action-angle variables associated with the transition state of the potential energy surface, as recognized by Miller.¹³ The bound degrees of freedom are quantized in the usual semiclassical [Einstein-Brillouin-Keller method (EBK)] fashion ($I_k = (n_k + \frac{1}{2})\hbar$; $k = 1, \dots, F - 1$), while the unbounded reaction coordinate is expressed as an imaginary action based on WKB theory ($I_F = \frac{i\theta}{\pi}$), with F being the total number of vibrational degrees of freedom. The tunneling (or transmission) probability is then expressed

as

$$P(E) = \frac{1}{1 + e^{2\theta}}, \quad (3)$$

and the CRP in turn is obtained by summing the probabilities over all states of the reactive complex,

$$G(E) = \sum_{n_k} \frac{1}{1 + e^{2\theta(n_k)}}, k = 0, \dots, F - 1. \quad (4)$$

It was established early on that SCTST gives the *exact* quantum result for the case of a parabolic barrier (i.e., inverted harmonic oscillator),¹³ although the most practically applicable form of SCTST appears in conjunction with second order vibrational perturbation theory (VPT2).¹⁴ The appeal of VPT2-SCTST stems from the fact that beyond the harmonic approximation, the only additional information that is required are the cubic, diagonal (*iii*), and semidiagonal (*ijj*) quartic force constants evaluated at the transition state geometry. Such a computation can be considered routine nowadays, given the rapid developments in computing power during the last couple of decades. The success of VPT2-SCTST has been registered on many occasions in the recent past in the form of various applications to compute reaction probabilities.^{15–18} For a detailed review of VPT2-SCTST, the reader is referred to a recent review by Nguyen *et al.*¹⁹ Along with its great success, the limitations of VPT2-SCTST have also been discussed in the literature, which occur when the region around the transition state is significantly anharmonic, or in the deep tunneling regime.²⁰ This has motivated the pursuit of improved semiclassical theories, such as the one proposed by Wagner.²¹

A logical but brute force approach to improve upon VPT2-SCTST is to move to VPT4-SCTST, the next nonvanishing order of perturbation theory in the energy expansion.²² A study of VPT4-SCTST for one-dimensional symmetric Eckart barrier was performed recently, where it was found to be in excellent agreement with the exact quantum mechanical result including in the deep tunneling regime (where it also offered substantial improvement over VPT2-SCTST). In this work, VPT4-SCTST is applied to a variety of one-dimensional potential barriers to further investigate its performance and to evaluate whether the findings here corroborate or contradict the previous results obtained for the symmetric Eckart barrier. There have been some attempts in the context of canonical vibrational perturbation theory (CVPT) to improve the SCTST rate.²³

Another important issue that arises in the application of SCTST based on VPT is the convergence of result with respect to the order of perturbation theory. An interesting question to ask in this regard is at which point can it be said that the converged result has been obtained and one does not need to move to the next order of perturbation theory? While one might assume that VPT4 achieves convergence for almost all practical purposes, a systematic study is nearly impossible to perform as the anharmonic constants beyond VPT4 (which, it should be noted, is also a recent development) are simply not available and would be extremely challenging to obtain, to say the least. Given this scenario, an interesting realization takes place that the complex-valued energies in the transition state region given by VPT [Eq. (5)] are an approximation to Siegert

eigenvalues, which are the corresponding *exact* energies and are obtained as the outgoing wave solutions of the Schrödinger equation, as was noted by Seideman and Miller.³ If one has access to Siegert eigenvalues that can be expressed as a Taylor series expansion in $(n + 1/2)$, then the convergence properties of SCTST can be analyzed in terms of the convergence of such an energy expression. However, one must take a note of the fact that these coefficients are generally not what one would get from VPT at the same order in $(n + 1/2)$. Nonetheless, such an analysis would hint at what one can expect in general for SCTST convergence based on a VPT approximation. Such an analysis is performed for the Morse as well as the asymmetric Eckart barrier and is one of the main results of this work.

In addition to the Morse and asymmetric Eckart barrier, the third example considered is the “adjacent well” barrier. It is a relatively little-known potential which has an analytic solution available for the transmission probability. It also represents a qualitatively realistic scenario in chemical reaction dynamics as it contains a bound well next to the barrier (hence the name). The paper is organized as follows: Sec. II briefly introduces the VPT-SCTST formalism. In Sec. III, the above mentioned model systems are studied based on the theory presented in Sec. II. The conclusions and future work are discussed in Sec. IV.

II. OVERVIEW OF VPT AND SCTST

The discussion in this section closely follows the pattern of previous study of VPT4-SCTST for the symmetric Eckart barrier, although some new comments are made here. The goal of vibrational perturbation theory is to obtain the vibrational energy levels of a system governed by the Watson Hamiltonian.²⁴ The energy is expressed as a polynomial in the $(n_k + \frac{1}{2})$, where n_k is the quantum number of the k th mode corresponding to the zeroth order harmonic oscillator Hamiltonian, solvable exactly. The energy polynomial can be expressed as follows:

$$E = V_0 + G + \sum_k W_k(n_k + \frac{1}{2}) + \sum_{kl} X_{kl}(n_k + \frac{1}{2})(n_l + \frac{1}{2}) + \sum_{klm} Y_{klm}(n_k + \frac{1}{2})(n_l + \frac{1}{2})(n_m + \frac{1}{2}) + \sum_{klmn} Z_{klmn}(n_k + \frac{1}{2})(n_l + \frac{1}{2})(n_m + \frac{1}{2})(n_n + \frac{1}{2}) + \dots \quad (5)$$

In the above expression, V_0 is the value of potential energy at the stationary point on the potential energy surface with respect to which the Taylor expansion has been carried out. The zeroth order energies, corresponding to VPT0, comprise V_0 and linear coefficients W_k (harmonic oscillator). The second order expansion, corresponding to VPT2, also includes terms G and X , while in fourth order expansion corresponding to VPT4, the term Y is added, along with additional contribution to the term W . The characteristic pattern of these anharmonic constants was discussed in some detail in the previous work. The VPT2 expressions for G and X are well-known in the literature¹⁴ and have been used widely both in the context of spectroscopy and computing reaction rates. A closed form

algebraic expression for Y_{ijk} , and fourth order contribution to W_k , has been derived in our group.²⁵ The analytic expressions for a one-dimensional case (which applies to the present work) are available and the corresponding equations are given as follows:

$$\begin{aligned} W &= \tilde{v}_e - \frac{67\phi_4^2}{9216\tilde{v}_e} + \frac{5\phi_6}{1152} + \frac{17\phi_3^2\phi_4}{512\tilde{v}_e^2} - \frac{19\phi_3\phi_5}{1152\tilde{v}_e} - \frac{385\phi_3^4}{27648\tilde{v}_e^3}, \\ X &= \frac{\phi_4}{16} - \frac{5\phi_3^2}{48\tilde{v}_e}, \\ G &= \frac{\phi_4}{64} - \frac{7\phi_3^2}{576\tilde{v}_e}, \\ Y &= \frac{\phi_6}{288} - \frac{7\phi_3\phi_5}{288\tilde{v}_e} + \frac{25\phi_3^2\phi_4}{384\tilde{v}_e^2} - \frac{17\phi_4^2}{2304\tilde{v}_e} - \frac{235\phi_3^4}{6912\tilde{v}_e^3}. \end{aligned} \quad (6)$$

It is to be noted that the equation for W_k was incomplete in the previous work, although it did not have any impact on the reported results since the missing terms were identically zero for the symmetric Eckart barrier. As noted in the previous work, the ϕ_n are the n th order derivatives of the energy in wavenumber with respect to the dimensionless normal coordinates q , which are defined in atomic units as follows:

$$\phi_n^{(a.u.)} = \frac{f_n^{(a.u.)}}{(2\pi c)^{n+2/2} m^{n/2} \tilde{v}_e^{n/2}}, \quad (7)$$

where $c = \alpha^{-1}$, with α being the fine structure constant, $\tilde{v}_e = \phi_2$, and f_n are the geometric derivatives of the potential [$f_n \equiv (\frac{d^n V}{dx^n})$]. Since $\tilde{v}_e$ is imaginary for the reaction coordinate, these derivatives can be real, imaginary, or half-imaginary depending on the order of derivative, n . This in turn governs whether the anharmonic constants are real or imaginary, based on the definitions given in Eq. (6).

An interesting observation is that if the anharmonic constants are evaluated as described above and substituted directly into Eq. (5), the resulting energies are complex valued. As noted earlier in the Introduction, these complex-valued energies are an approximation to the *true* energies obtained by solving the Schrödinger equation for the transition state, which are known as *Siebert eigenvalues*. The latter are defined as the outgoing wave solution to the Schrödinger equation for an unbound system and can be used to evaluate tunneling probabilities semiclassically²⁶ as well as quantum mechanically.³ It should also be noted that once the semiclassical transformation— $(n_k + \frac{1}{2}) \rightarrow \frac{i\theta}{\pi}$ —has been carried out for the reaction coordinate, the VPT energies become real valued. This transformation reminds one of the *Wick rotation* performed in quantum statistical mechanics to go from real time (Minkowski space) to imaginary time (Euclidean space). While a detailed discussion of this is beyond the scope of (and not directly related to) this work, the interested reader might look into the derivation of the WKB approximation in the framework of path integrals.^{27,28}

Once the anharmonic constants have been obtained and the semiclassical transformation has been performed, a simple quadratic (for VPT2) or cubic (for VPT4) equation results which can be solved for θ , which in turn yield the $P(E)$ based

on Eq. (3). For VPT2, the corresponding energy equation is

$$E = V_0 + G - W \frac{\theta}{\pi} - X \frac{\theta^2}{\pi^2}, \quad (8)$$

which has the solutions

$$\theta_{\pm}^{VPT2} = -\frac{\pi\omega}{2X} \left\{ 1 \pm \left(\sqrt{1 - \frac{4X\Delta E}{\omega^2}} \right) \right\}, \quad (9)$$

where $\Delta E = E - V_0 - G$ and $W = \omega$ is the barrier frequency. For VPT4, the corresponding energy equation is

$$E = V_0 + G - W \frac{\theta}{\pi} - X \frac{\theta^2}{\pi^2} + Y \frac{\theta^3}{\pi^3}. \quad (10)$$

An analytic solution to the above cubic equation can be obtained but is not reported here for brevity. In addition, the *correct* root needs to be followed for both VPT2 and VPT4 with respect to the change in energy, E . For the examples considered in this work, root switching was not encountered although a possibility of it happening cannot be ruled out. Before discussing the results, it should be noted that the failure of VPT2 based on the solution of Eq. (8) has been discussed in detail in previous studies.^{19,21} These are situations when no real roots can be found for θ , based on Eq. (9). By contrast, for VPT4, since all the coefficients of the cubic equation are real, there has to be one real root amongst the three. Therefore, the VPT4 equation always has a solution, although it might still be far away from the exact semiclassical solution of θ . To summarize this section, the procedure for computing $P(E)$ is as follows: first the transition state (which is the same as the barrier maximum for one-dimensional systems) is located for the potential barrier. Next, up to sixth derivatives of the potential energy are evaluated at this point. Then the anharmonic constants are evaluated based on Eq. (6). Equations (8) and (10) are then solved to obtain θ . Finally, the transmission coefficient is obtained using Eq. (3). In Sec. III, the performance of VPT2- and VPT4-SCTST will be discussed through the means of model one-dimensional potential barriers, with an emphasis on the deep tunneling regime.

III. MODEL SYSTEM STUDIES: VPT4-SCTST

In this section, tunneling probabilities are computed for four different one-dimensional model systems: (i) Morse barrier, (ii) asymmetric Eckart barrier, (iii) adjacent well barrier, and (iv) first order approximation to the symmetric Eckart barrier, based on the equations presented in Sec. II. All the barriers are characterized by three parameters: height of the barrier (V_0), width of the barrier (a), and mass of the particle (m). In the case of the asymmetric Eckart barrier, there is an additional asymmetry parameter (μ) which governs the difference between the reactant and product asymptote. Unless otherwise stated, all the parameters are chosen to mimic the behavior of the collinear H + H₂ reaction ($V_0 = 0.015\,619\,3$, $a = 0.734$, $m = 1060.0$; all in atomic units), as was done in the previous study. All the anharmonic constants are presented in the units of $\frac{1}{2\pi c}$ as this makes the algebraic analysis easier.

A. Morse barrier

In addition to the harmonic oscillator, the potential that has received the most attention in chemical physics is probably the Morse potential. The success of this potential in diatomic spectroscopy as well as in treating anharmonicity for bond stretching in polyatomic molecules is unprecedented. Arguably, this cannot be said for its inverted counterpart, the Morse barrier (Fig. 1), in the context of the barrier penetration problems (for example, tunneling). The fact that VPT2 gives exact eigenvalues for the bound Morse potential has been noted well in the literature; however, a clear discussion of VPT2-SCTST for the Morse barrier, although trivial, has not been reported before (to the best of our knowledge). Below, the analytic anharmonic constants for the Morse barrier are presented based on Eq. (6), and the tunneling probability, $P(E)$, is computed based on these expressions. The Morse barrier is defined as follows:

$$V(x) = V_0 (2e^{\frac{x}{a}} - e^{\frac{2x}{a}}). \quad (11)$$

The VPT anharmonic constants up to fourth order are obtained as follows:

$$\begin{aligned} W &= \sqrt{\frac{2V_0}{ma^2}}, \\ X &= -\frac{1}{2ma^2}, \\ G &= 0, \\ Y &= 0. \end{aligned} \quad (12)$$

The action variable theta and the corresponding $P(E)$ are obtained as follows:

$$\theta_{VPT2} = \pi(\sqrt{V_0/\Delta} - \sqrt{E/\Delta}), \quad (13)$$

$$P(E) = \frac{1}{1 + e^{2\pi(\sqrt{V_0/\Delta} - \sqrt{E/\Delta})}}, \quad (14)$$

where $\Delta = \hbar^2/2ma^2$.

It is to be noted that this coincides with the result one obtains from WKB theory for the Morse barrier. In addition, since Y and higher order contribution to W are zero, VPT4-SCTST will give the same $P(E)$ as that of VPT2-SCTST. What is perhaps more interesting to note is that this is also the result obtained by an *exact* semiclassical treatment. The latter can be easily computed based on the Siegert eigenvalues of the Morse barrier,²⁹ which can be expressed based

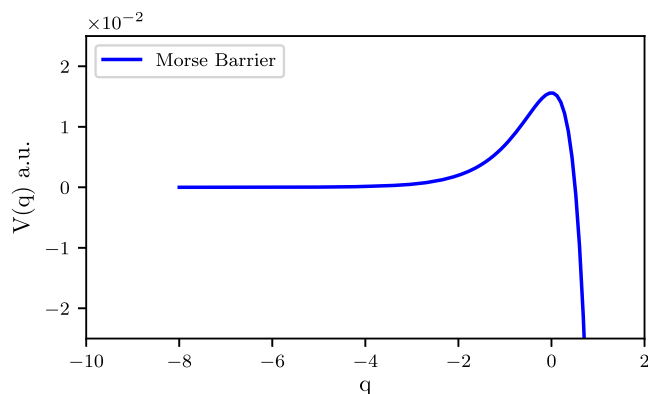


FIG. 1. Morse barrier.

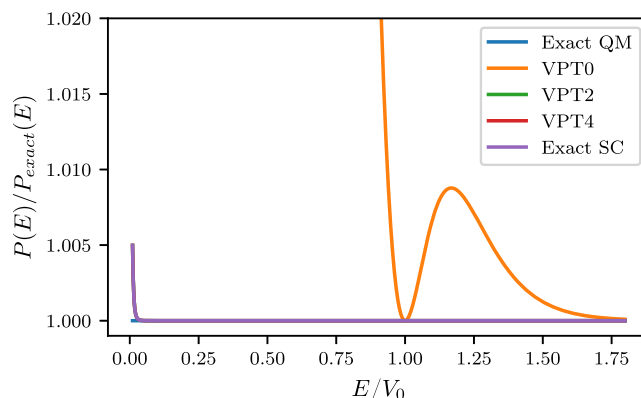


FIG. 2. Ratio of barrier transmission coefficients computed by VPT-SCTST approaches to the corresponding exact quantum result for the Morse barrier, covering a wide range of energies below and above the barrier. It is to be noted that VPT2 = VPT4 = Exact SC (overlapping curves).

on the analytic continuation of their bound counterparts as follows:³⁰

$$\epsilon = V_0 \left[1 - i(n + \frac{1}{2}) \frac{1}{a\sqrt{2mV_0}} \right]^2. \quad (15)$$

Expanding the above expression in $(n + \frac{1}{2})$ gives the same expansion coefficients as in Eq. (12) obtained from VPT, thus giving the same θ , and $P(E)$, as VPT2-SCTST. These tunneling probabilities are plotted as a ratio to the exact quantum mechanical $P(E)$ in Fig. 2. Another interesting caveat that seems to go unnoticed is that the above $P(E)$, which corresponds to VPT4-SCTST = VPT2-SCTST = WKB = Exact Semiclassical, is still not equal to the exact quantum mechanical tunneling probability for the Morse barrier, which is given as follows:³¹

$$P_{exact}(E) = \frac{1 - e^{-4\pi(\sqrt{E/\Delta})}}{1 + e^{2\pi(\sqrt{V_0/\Delta} - \sqrt{E/\Delta})}}. \quad (16)$$

Despite the subtle difference between the exact semiclassical and the exact quantum mechanical result, it is emphasized that for all practical purposes $P(E)$ given by VPT2-SCTST serves as the exact result. This is seen in Fig. 3, where the $P(E)$ are plotted in the deep tunneling regime. The VPT2-SCTST

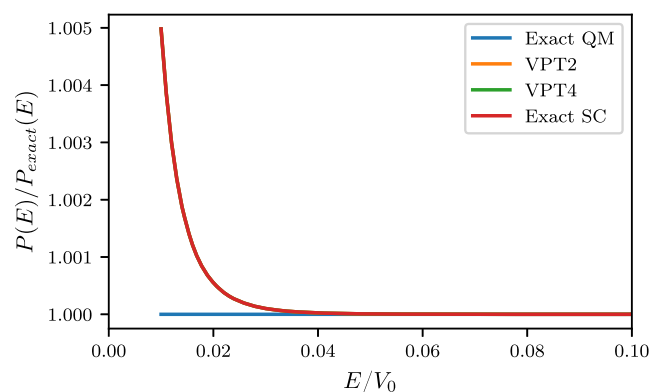


FIG. 3. Similar to Fig. 2, but displaying the deep tunneling regime.

$P(E)$ overlaps the exact QM result up to about 4% below the barrier height.

B. Asymmetric Eckart barrier

The next model to be discussed is the asymmetric Eckart barrier (Fig. 4), which is a realistic model for the reaction coordinate for many chemical reactions. The symmetric Eckart barrier, which was the subject of previous study, becomes a special case of this model. It is of interest to study the behavior of VPT4-SCTST for this barrier and to explore whether the excellent performance of the former for the symmetric case also holds here. Before discussing the results, a comment on the mathematical form of potential is in place. The most commonly used definition of the potential for the asymmetric Eckart barrier is as follows:³²

$$V(y) = \frac{Ay}{1+y} + \frac{By}{(1+y)^2}, \quad (17)$$

where $y = e^{\frac{2\pi(x-x_0)}{L}}$ and $x_0 = \frac{\mu L}{\pi}$, the location of barrier maximum. However, in this work, an alternative form used by Ryaboy and Moiseyev²⁶ is employed, which is expressed as follows:

$$V(x) = -V_0 \cosh^2(\mu) [\tanh[(x - \mu a)/a] + \tanh(\mu)]^2 + V_0 e^{-2\mu}. \quad (18)$$

The two forms are equivalent, and one can easily map the parameters of one onto the other. The Ryaboy and Moiseyev form is preferred here as the analytical Siegert eigenvalues and the exact semiclassical $P(E)$ reported by them are used here, where the latter is given as follows:

$$P_{SC}(E) = \frac{1}{1 + e^{\pi[\omega_{rm} - \sqrt{\alpha E(1 + \sqrt{1 + 2(V_0/E) \sinh(2\mu)})}]}}}, \quad (19)$$

where $\alpha = \frac{2ma^2}{\hbar^2}$ and $\omega_{rm} = \sqrt{4\alpha V_0 \cosh^2(\mu) - 1}$. It is to be noted that the expression reported in their work contains a typo where E is missing from V_0/E . For $\mu = 0$, the potential becomes the symmetric Eckart Barrier. Next, the anharmonic constants for the asymmetric Eckart barrier up to fourth order are obtained analytically, after some laborious algebraic manipulation, as follows:

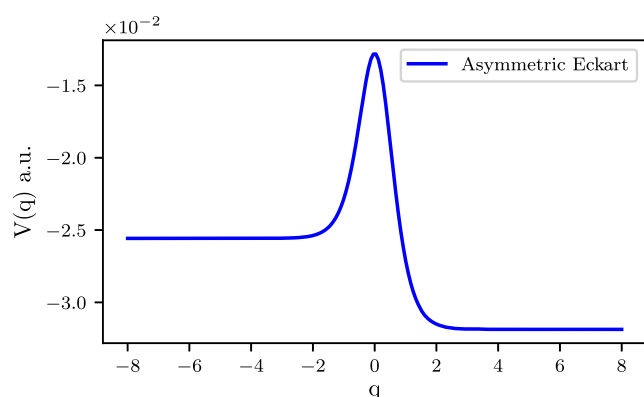


FIG. 4. Asymmetric Eckart barrier.

$$\begin{aligned} G &= -\frac{1}{4\alpha} \operatorname{sech}^2(\mu), \\ X &= \frac{1}{2} \left\{ \frac{7 \cosh^2(\mu) - 15 \sinh^2(\mu) - 9}{\alpha \cosh^2(\mu)} \right\}, \\ Y &= \left\{ \frac{288 \sinh^2(\mu) - 5670 \sinh^4(\mu)}{576} \right\} \frac{\omega_b^3}{V_0^2}, \\ W &= \omega_b + \left\{ \frac{216 - 288 \cosh^2(\mu)}{2304} \right\} \frac{\omega_b^3}{V_0^2}, \end{aligned} \quad (20)$$

where $\alpha = \frac{2ma^2}{\hbar^2}$ and $\omega_b = \sqrt{\frac{2V_0}{ma^2 \cosh^2(\mu)}}$, the barrier frequency.

An evident difference to that of the symmetric case is that the fourth order constant, Y , is non-zero. It is also easily verified that for $\mu = 0$, all the constants reduce to the case of the symmetric Eckart barrier.

Tunneling probabilities are plotted in Fig. 5 for $\mu = 0.1$. Although VPT4-SCTST does better than its VPT2 counterpart, the performance of VPT4-SCTST in this case is not as impressive relative to that of the symmetric case. The tunneling correction term obtained for the symmetric case in the previous study is applied to this barrier, which does improve the VPT2 result but remains somewhat less accurate than the VPT4 result. This behavior is plotted in Fig. 6 for the deep tunneling regime. These results for the asymmetric Eckart barrier indicate the importance of higher order terms in the VPT expansion of energy. While this is partly hinted by a non-zero value of the anharmonic constant Y , it is also of interest to know the importance of higher order terms beyond the fourth order to gain insight into the convergence properties of VPT in the context of SCTST. In addition, it is desirable to obtain a correction term tailored specifically for this barrier in the same spirit as it was done for the symmetric case, which will ideally perform better than the correction term for the symmetric Eckart barrier.

As noted earlier in the Introduction, an insight into both of these issues can be obtained by analyzing the Siegert eigenvalues of this barrier, which are expressed as follows:²⁶

$$\epsilon = \alpha^{-1} \left[-2\beta_2 + (\beta_1 + i(n + 1/2))^2 + \frac{\beta_2^2}{(\beta_1 + i(n + 1/2))^2} \right], \quad (21)$$

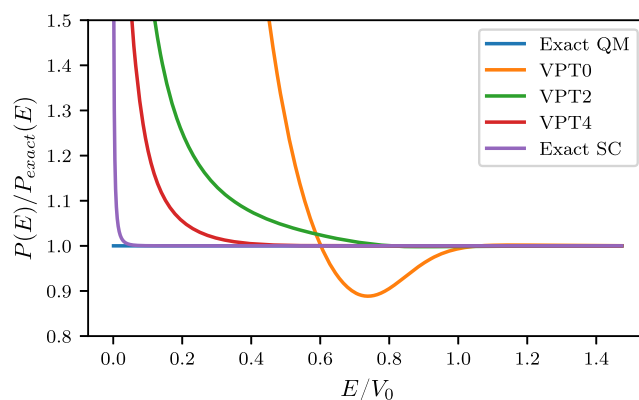


FIG. 5. Ratio of barrier transmission coefficients computed by VPT-SCTST approaches to the corresponding exact quantum result for the asymmetric Eckart barrier, covering a wide range of energies below and above the barrier.

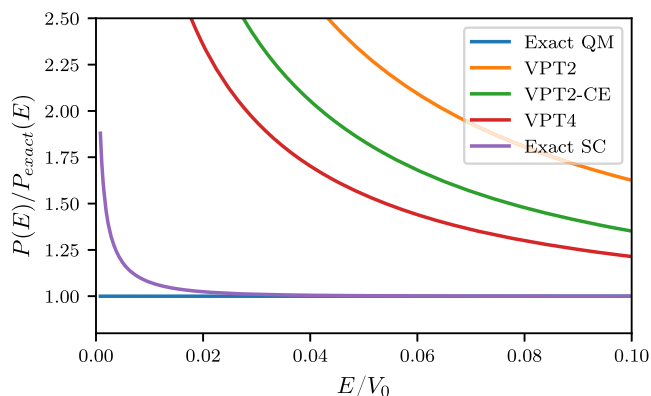


FIG. 6. Similar to Fig. 5, but displaying the deep tunneling regime.

where $\beta_1 = \sqrt{\alpha V_0 \cosh^2(\mu) - 1/4}$ and $\beta_2 = \frac{\alpha V_0 \sinh(2\mu)}{2}$ and α is given as before. Expanding this series in $(n + \frac{1}{2})$ gives the following expansion coefficients up to fourth order:

$$\begin{aligned} G &= \frac{1}{\alpha} \left\{ \beta_1^2 - 2\beta_2 + \frac{\beta_2^2}{\beta_1^2} \right\}, \\ X &= -\frac{1}{\alpha} \left\{ 1 + \frac{3\beta_2^3}{\beta_1^4} \right\}, \\ Y &= \frac{4\beta_2^2}{\beta_1^5}, \\ W &= \frac{2}{\alpha} \left\{ \beta_1 - \frac{\beta_2^2}{\beta_1^3} \right\}. \end{aligned} \quad (22)$$

The above expressions for the anharmonic constants can be expressed directly in terms of potential parameters, but a closed form expression similar to those presented in Eq. (20) has eluded us. This makes a direct, side-by-side comparison with VPT4 anharmonic constants a difficult task. A numerical comparison is performed in Fig. 7, where the difference between VPT and exact semiclassical expressions for the four terms has been plotted as a function of the asymmetry parameter μ . As one would expect, for the symmetric case ($\mu = 0$), X , G , and Y ($=0$) are equal for Siegert eigenvalues and VPT, while for any $\mu > 0$, this is not the case. The earlier seen Morse barrier was an exception to this general rule where all the coefficients happen to be equal, while the difference in the case of the symmetric Eckart barrier was also minor, only appearing in the term W .³³ Given this situation for the asymmetric Eckart Barrier, it seems implausible, if not impossible, to obtain a correction term specific to its own case.

It is to be emphasized again that the energies obtained by the VPT expansion [Eq. (5)] are an approximation to the true Siegert eigenvalues, which are the outgoing wave solution to the Schrödinger equation at the transition state geometry, as mentioned earlier. A similar structure of VPT energies to the true Siegert eigenvalues will indicate a faster convergence of the former. This was the case with the Morse barrier and to large extent for the symmetric Eckart barrier as well. Here, Taylor expansion of the Siegert eigenvalues confirms the presence of non-zero higher order terms, which are not accounted for even by VPT4, certainly not by VPT2. The expansion of the exact Siegert eigenvalue expression given by Ryaboy and Moiseyev in terms of $(n + 1/2)$ is non-terminating, and the existence

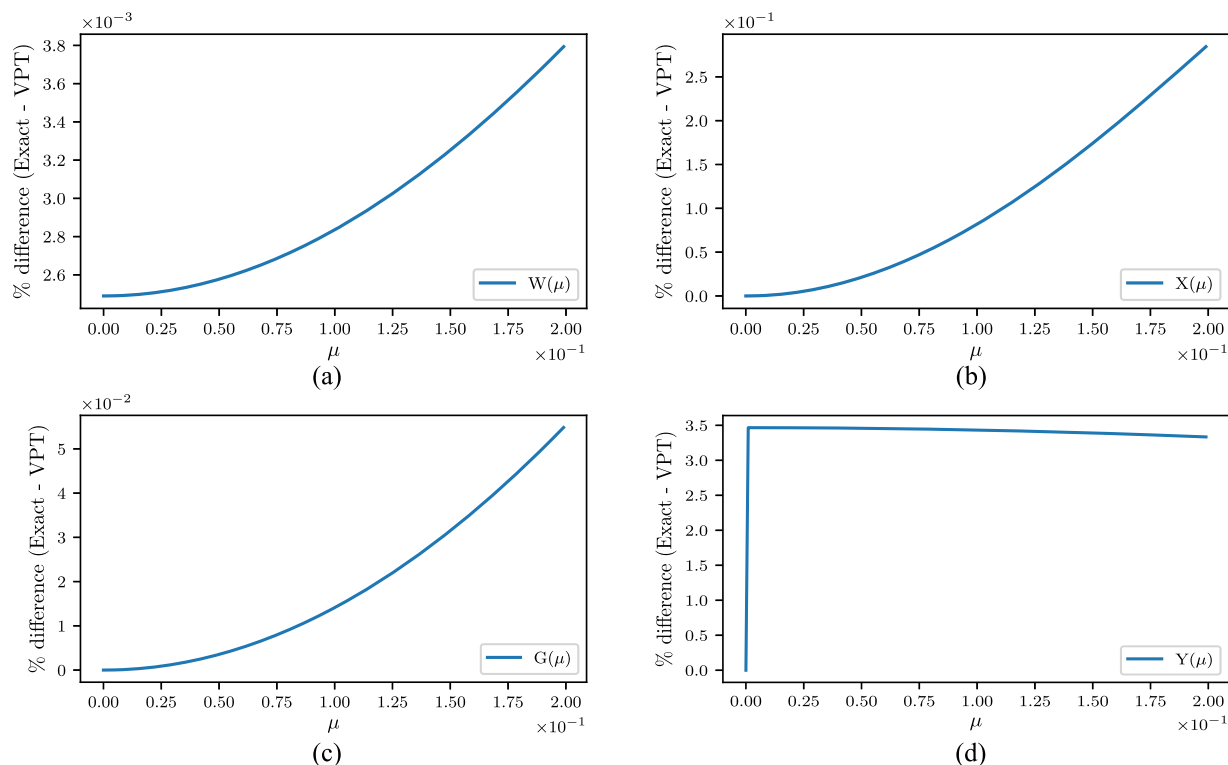


FIG. 7. Percent difference between the anharmonic constant as obtained by Taylor expansion of Siegert eigenvalues (Exact) to those obtained by VPT4. The $\mu = 0$ case corresponds to the symmetric Eckart barrier, where except W all other constants are the same in both cases.

of all terms beyond cubic clearly shows that VPT4-SCTST will miss contributions to $P(E)$. To gauge the importance of these higher order terms, $P(E)$ calculated from the Siegert expansion via the semiclassical transformation are shown in Fig. 8. It is clear that the cubic approximation is far from convergence in this sense, which provides some evidence that contributions from sixth-order VPT (and beyond) are numerically significant.

C. Adjacent well barrier

The number of model barrier potentials for which the Schrödinger equation can be solved exactly is limited. One such potential is given as follows, for which the tunneling probability has been obtained analytically. The potential barrier is defined as follows:

$$V(x) = V_0 \tanh\left(\frac{x}{a}\right) \operatorname{sech}\left(\frac{x}{a}\right), \quad (23)$$

while the exact quantum mechanical $P(E)$ for this barrier is given as follows:³⁴

$$P(E) = \frac{\sinh^2(2\pi k)}{[(\cosh(2\pi k) + \cos(2\pi\alpha))(\cosh(2\pi k) + \cosh(2\pi\beta))]}, \quad (24)$$

where $k = \sqrt{E/\Delta}$, $\alpha = \operatorname{Re}(\sqrt{1/4 + iV_0/\Delta})$, $\beta = \operatorname{Im}(\sqrt{1/4 + iV_0/\Delta})$, and $\Delta = \hbar^2/(2ma^2)$.

This potential barrier has been plotted in Fig. 9 for the same value of potential parameters (V_0, a) as for the previous examples. As noted earlier in the Introduction, this is a qualitatively plausible representation of the reaction coordinate for many chemical reactions, when there is an adjacent well to the barrier in consideration. Surprisingly, this model potential has received little to no attention in the literature (to the best of our knowledge). The analysis of tunneling probabilities and the correction term is performed as for the previous examples. The analytic anharmonic constants up to the fourth order are obtained as follows:

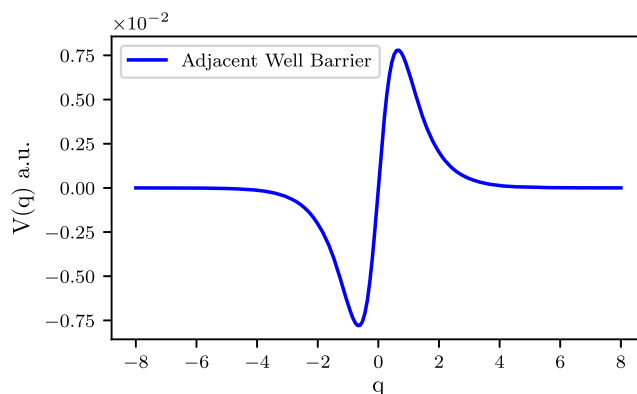


FIG. 9. Adjacent well barrier.

$$\begin{aligned} W &= \omega_b - \frac{\omega_b^3}{16V_0^2}, \\ X &= -\frac{1}{2ma^2}, \\ G &= -\frac{1}{8ma^2}, \\ Y &= 0. \end{aligned} \quad (25)$$

The ratio of tunneling probabilities is plotted in Fig. 10 for energies below and above the barrier. The performance of VPT4-SCTST is excellent in the deep tunneling regime, almost mimicking the behavior of the symmetric Eckart barrier. This should not come as a surprise when one looks at the structure of anharmonic constants obtained above, which have an uncanny resemblance to the symmetric Eckart barrier case. What is rather counter-intuitive is the effect of the previously obtained correction term, as the tunneling probability behaves in a completely nonphysical manner when the correction term is applied to the VPT2-SCTST result. This behavior can be seen in Fig. 11(a). It shows that the correction term can be quite sensitive, owing to even a small difference in the definition of anharmonic constants. Interestingly, it is possible to obtain a correction term specific to this case, as the only difference between the VPT2 and VPT4 expression appears in the term W , in the same spirit to that of the symmetric Eckart barrier.³³ The new correction term is given as

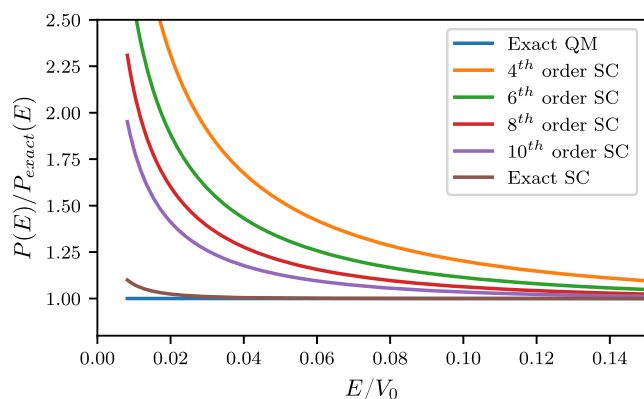


FIG. 8. Ratio of barrier transmission coefficients computed using the anharmonic constants obtained by a Taylor series expansion of Siegert eigenvalues (n th order SC, see Sec. III B) to the corresponding exact quantum result for the asymmetric Eckart barrier in the deep tunneling regime.

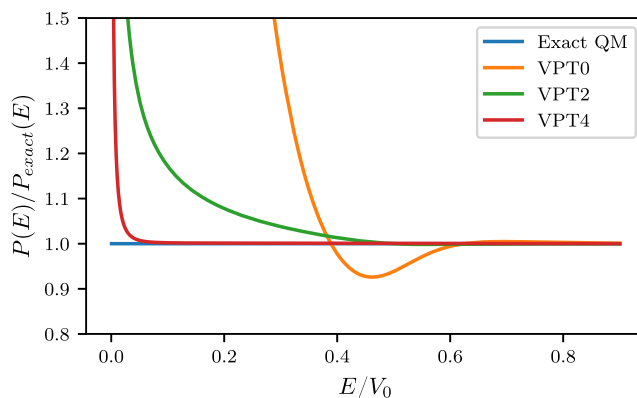


FIG. 10. Ratio of barrier transmission coefficients computed by VPT-SCTST approaches to the corresponding exact quantum result for the adjacent well barrier, covering a wide range of energies below and above the barrier.

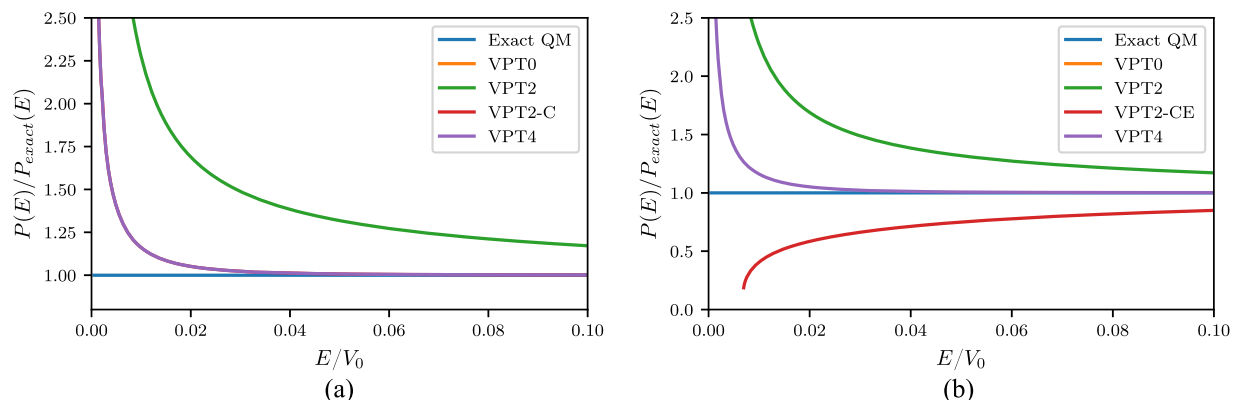


FIG. 11. Ratio of tunneling probabilities for the adjacent well barrier in the deep tunneling regime: (a) with the symmetric Eckart correction term applied to VPT2 and (b) with its own custom correction term applied to VPT2. The latter reproduces the VPT4-SCTST result as the two curves are seen overlapping (violet and red).

follows:

$$\gamma = 1 - \frac{X^2}{4\omega_b^2}. \quad (26)$$

When one scales the VPT2 frequency ω_b with the new correction term ($\tilde{\omega}_b = \gamma\omega_b$), the new VPT2-SCTST $P(E)$ equals to that of VPT4-SCTST $P(E)$. This behavior can be seen in Fig. 11(b) where the corrected VPT2 curve and VPT4 curve are on top of each other. It is also emphasized that this correction term is based on the difference between VPT2 and the *actual* VPT4 anharmonic constants and not on the difference between VPT2 and terms obtained by a Taylor series expansion of Siegert eigenvalues, as the latter are not available in this case.³⁵ In this sense, this correction term is equivalent to the *first* correction term of the previous work. The difference between the two correction terms (symmetric Eckart and the current barrier) is subtle, differing only by a factor of 1/2, but it would be interesting to find systems where the current term works well in place of the symmetric Eckart one.

D. First order approximation to the symmetric Eckart barrier

An interesting one-dimensional example is obtained by truncating the Taylor expansion of the symmetric Eckart barrier after first order. The resulting barrier mimics the symmetric Eckart barrier in the vicinity of the transition state region but drops off to zero rather slowly. The two potentials are shown in Fig. 12 for comparison. The first order barrier does not have an analytic solution, but the quantum mechanical $P(E)$ can be obtained numerically exactly by any suitable grid based method. The potential barrier is defined as follows:

$$V(x) = \frac{V_0}{x^2 + a^2}. \quad (27)$$

The corresponding anharmonic constants are defined as follows:

$$\begin{aligned} X &= -\frac{3}{4ma^2}, \\ G &= -\frac{3}{16ma^2}, \\ W &= \omega_b - \frac{33\omega_b^3}{256V_0^2}, \\ Y &= -\frac{3\omega_b^3}{64V_0^2}, \end{aligned} \quad (28)$$

where ω_b is the barrier frequency. The structure of constants X , G , and W is essentially similar to the symmetric Eckart barrier case, as perhaps is to be expected. However, the fourth order constant Y is non-zero in this case. In order to compare tunneling probabilities, VPT2 and VPT4-SCTST $P(E)$ are obtained as for the previous cases. Because of numerical issues, a direct comparison of $P(E)$ is made this time, instead of plotting ratios as was done in the previous examples. In Fig. 13, for $E > 0.4V_0$, i.e., for energies above roughly half of the barrier height, the usual pattern is followed where VPT4 and the VPT2 with the symmetric Eckart correction (VPT2-C) give more accurate $P(E)$ compared to the VPT2-SCTST. In Fig. 13, for energies $E \approx < 0.3125V_0$, VPT2 ceases to have a solution, which can be proved by analyzing the roots of quadratic equation (8). This is an interesting scenario because in this case even the correction term does not

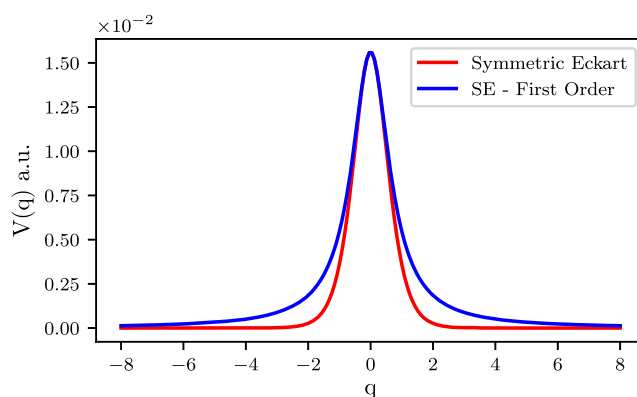


FIG. 12. First order approximation to the symmetric Eckart barrier.

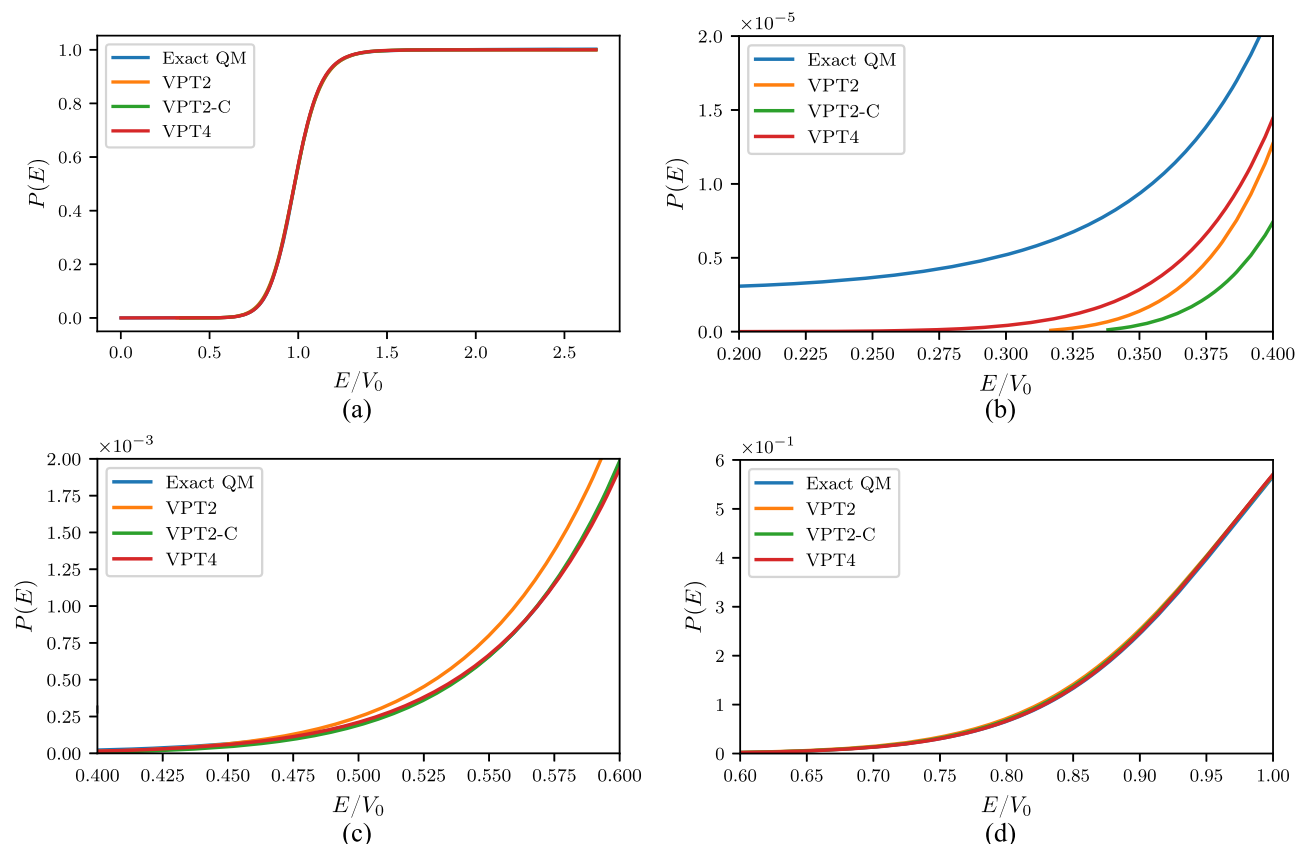


FIG. 13. Tunneling probabilities for the first order approximation to the symmetric Eckart barrier. VPT2-SCTST does not have any solution for $E < 0.3125V_0$, as seen in (b).

lead to an improved solution. In addition, the performance of VPT4 also drops off significantly in this energy range. This is indicative of the importance of higher order terms. A detailed convergence analysis cannot be performed here due to the lack of knowledge of analytic Siegert eigenvalues, but this example once again highlights the limitations of VPT2-SCTST.

IV. CONCLUSIONS

In summary, the performance of VPT4-SCTST to compute tunneling probabilities has been studied by means of a variety of one-dimensional potential barriers. The examples are selected for which exact analytic results are available for the tunneling probabilities (except the last example). One reason for this choice is the focus on comparison in the very deep tunneling regime, where numerics become extremely sensitive [$P(E)$, for example, is often of order 10^{-9} or less]. It must be emphasized that the contribution of this region to the CRP in practical rate calculations is negligible in such cases, and threshold tunneling is the most important in such a scenario. However, the primary aim of this study is to investigate the convergence properties of VPT-SCTST.

A number of interesting results have been reported. First and foremost, the Morse barrier has been revisited and it is shown that even though VPT2 captures the correct energy expansion at the transition state geometry (in the same spirit as for its bound sibling), the SCTST tunneling probability is

not equal to its exact quantum mechanical counterpart. This (minuscule) difference is attributed to the semiclassical limitation of the model, as VPT2-SCTST is equal to the exact semiclassical result, where the latter has been obtained using the Siegert eigenvalues of the Morse barrier.

The behavior of the asymmetric Eckart barrier with respect to VPT4-SCTST has turned out to be somewhat different from its symmetric counterpart, as in the deep tunneling regime the performance is clearly not as good. It is noted, via a Taylor series expansion of its analytic Siegert eigenvalues, that terms beyond fourth order in the energy expansion (and in computing tunneling probability) are important in the deep tunneling regime. The correction term obtained for the symmetric Eckart barrier in the previous study improves the VPT2-SCTST result, although a modified correction term tailored specifically to the asymmetric case could not be found due to a complex structure of its anharmonic constants. Another variant of the symmetric Eckart barrier, its first order approximation, is indicative of a similar behavior, as the fourth order term here is non-zero. The comparison and analysis in this case leave a lot to be desired due to a lack of unavailability of analytic results.

The adjacent well barrier is another interesting model system where an analytically exact tunneling probability is available. The performance of VPT4-SCTST in this case is excellent, essentially the same as the symmetric Eckart barrier case, which can be attributed to the similar structure of their anharmonic constants. What is interesting is the fact

that despite having almost identical structure of W , G , X , Y , the correction term for symmetric Eckart failed completely in this case. Fortunately, owing to the same analysis performed for the symmetric Eckart case, a new correction term has been obtained [Eq. (26)] for this barrier, which could be tested for real systems. The takeaway is also that the previously obtained correction term can be sensitive to subtle changes in the shape of the potential and must be used with care.

As mentioned earlier, the full derivation and implementation of anharmonic constants for polyatomic molecules are complete in our group and we look forward to applying to VPT4-SCTST methodology to a real system in the foreseeable future. Another intriguing aspect that is currently under investigation in our group is the physical interpretation of Siegert eigenvalues in the context of passage through a barrier and their role in obtaining a quantum mechanical $P(E)$. While such an analysis was first discussed by Seideman and Miller³ and later applied by Ryaboy and Moiseyev for the asymmetric Eckart barrier,²⁶ the potential of this approach and understanding of Siegert states remain incomplete.

ACKNOWLEDGMENTS

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³⁵One could attempt to obtain analytic Siegert eigenvalues for this barrier, which can then be used to find a correction term in the same spirit to that of the symmetric Eckart barrier. It might be useful for other purposes as well (see the main text).